BarrierLens — Stage 2 Modelling Report

Project P48 | Generated 2026-07-25

1. Stage 2 Objective

Predict downstream maternal healthcare utilisation (ANC care gap) and family planning access (unmet need) using socioeconomic features, Stage 1 out-of-fold barrier probabilities, engineered vulnerability indices, and woman-level risk clusters.

2. Dataset and Analytical Sample

NFHS-5 individual-level extract (724,115 women). Each target uses its own restricted sample where structural missingness applies — targets are not imputed.

3. Target Definitions

target_anc_gap: 1 if m14 < 4 ANC visits, else 0 (requires m14 in raw extract). target_unmet_fp: 1 for 'unmet need for spacing/limiting', 0 otherwise among valid v626a categories. Leakage-prone s245a/b/h excluded.

4. Data Preprocessing

Stage 1 features + 3-fold OOF barrier probabilities + composite score + MiniBatchKMeans clusters (one-hot encoded). Per-target row filtering before split.

5. XGBoost Methodology

Separate XGBClassifier per target; tree_method='hist'; scale_pos_weight = neg/pos; 80/20 stratified hold-out; 3-fold CV stability check on training fold.

6. Hyperparameters

Tuned n_estimators (300–400) and max_depth (5–7) via GridSearchCV on stratified subsample; learning_rate=0.08, subsample=0.8, colsample_bytree=0.8.

Note: target_anc_gap was skipped — m14 is absent from the current 32-column NFHS5_Individual.csv extract (0 non-null rows). Re-pull m14 from IAIR7EFL.DTA to enable ANC gap modelling.

7. Evaluation Metrics

ROC-AUC and F1 prioritised over accuracy due to class imbalance.

8. XGBoost Performance

	Accuracy	ROC-AUC	Precision	Recall	F1-Score	TrainSize	TestSize	CV_ROC-AUC	Baseline_ROC-AUC	Full_ROC-AUC	Barri
met_fp	0.5927	0.668	0.1603	0.6675	0.2586	373487	93372	0.6652	0.6683	0.6662	-0.00

9. SHAP Analysis

Top SHAP features — Unmet Family Planning Need

feature	mean_abs_shap
v012	0.3195
v013	0.11006
v130_hindu	0.10945
v501_married	0.10903
v106	0.10264
media_exposure_index	0.09308
household_barrier_prob	0.06819
v481_yes	0.06274
composite_barrier_score	0.06273
logistic_barrier_prob	0.05532

10. Barrier Uplift Analysis

Target	Baseline_ROC-AUC	Full_ROC-AUC	Barrier_Uplift
target_unmet_fp	0.6683	0.6662	-0.0022

11. Model Comparison (LR vs RF vs XGBoost)

Model	Target	Accuracy	ROC-AUC	Precision	Recall	F1-Score	Baseline_ROC-AUC	Full_ROC-AUC	Barrier_Uplift
Logistic Regression	target_unmet_fp	0.5847	0.6591	0.1573	0.6665	0.2546	0.6566	0.6583	0.0017
Random Forest	target_unmet_fp	0.6041	0.6672	0.1615	0.6493	0.2587	0.6601	0.6572	-0.003
XGBoost	target_unmet_fp	0.5927	0.668	0.1603	0.6675	0.2586	0.6654	0.6618	-0.0036

12. Key Findings

• Unmet Family Planning Need: ROC-AUC=0.6680, barrier uplift=-0.0022.

• Best overall model for Unmet Family Planning Need: XGBoost (ROC-AUC=0.6680).

13. Limitations

• Current 32-column extract supports only ANC gap and unmet FP targets — not child nutrition, NCD, or cancer screening.

• target_anc_gap requires m14; if absent from the CSV/DTA extract, only unmet FP is modelled.

• Barrier probabilities are OOF estimates — uplift reflects predictive association, not causation.

• Single 80/20 hold-out; geographic/temporal drift not assessed.

14. Research Implications

If barrier-uplift is positive, Stage 1 barrier exposure adds predictive signal for downstream maternal and reproductive health outcomes beyond socioeconomic status alone — supporting integrated barrier-outcome modelling for policy targeting.

15. Conclusion

Stage 2 XGBoost models were trained on individual-level NFHS-5 data with strict leakage controls. Results above are taken directly from saved evaluation CSVs; see notebooks 11–12 for reproducible execution.